

LA GRANDE 4, QC, Canada

WMO: 710030

Lat: 53.755N

Lon: 73.675W

Elev: 306

StdP: 97.70

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.6	(c) -34.9	(d) -41.8	(e) 0.1	(f) -37.5	(g) -39.2	(h) 0.1	(i) -34.8	(j) 11.0	(k) -15.9	(l) 9.7	(m) -15.9	(n) 0.8	(o) 60	(p) 0.773

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	26.4	16.6	24.3	15.7	22.4	15.1	18.6	23.1	17.5	21.6	16.5	20.3	4.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.7	20.1	16.1	11.8	18.8	15.0	11.0	18.0	54.3	23.1	50.7	21.6	47.5	20.2	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.9	8.6	7.7	-43.2	30.1	3.1	1.8	-45.5	31.4	-47.3	32.4	-49.1	33.4	-51.3	34.7
			-43.2	21.1	2.9	1.8	-45.3	22.4	-47.0	23.5	-48.6	24.5	-50.7	25.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.0	-21.6	-20.7	-13.7	-4.4	5.0	11.9	14.8	13.6	8.7	2.4	-5.6	-15.2		
	DBStd	14.13	8.07	7.80	8.51	6.55	5.43	4.77	3.44	3.48	4.17	4.26	5.41	7.72		
	HDD10.0	4783	979	861	736	431	174	34	3	7	73	238	469	780		
	HDD18.3	7435	1238	1094	994	681	415	199	119	154	291	493	719	1038		
	CDD10.0	413	0	0	0	0	18	90	152	117	34	3	0	0		
	CDD18.3	23	0	0	0	0	1	6	9	6	1	0	0	0		
	CDH23.3	282	0	0	0	0	21	88	114	52	7	0	0	0		
	CDH26.7	50	0	0	0	0	3	20	20	7	1	0	0	0		
(14) Wind	WSAvg	3.5	3.2	3.2	3.3	3.5	3.5	3.5	3.4	3.5	3.9	3.7	3.8	3.3		
(15) Precipitation	PrecAvg	759	35	23	28	33	48	75	114	108	116	80	58	42		
	PrecMax	959	57	58	49	68	106	141	254	207	228	139	98	74		
	PrecMin	611	12	11	5	11	12	24	58	68	30	32	30	16		
	PrecStd	89	11	11	13	17	24	27	48	34	46	24	15	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.0	2.5	7.0	13.4	26.4	29.5	29.0	27.8	23.9	17.3	7.6	2.5		
		MCWB	0.7	0.7	3.6	6.6	14.3	19.0	17.7	18.0	17.5	13.1	6.5	1.7		
	2%	DB	-2.3	-0.9	3.7	9.7	21.9	25.7	26.3	24.8	20.5	13.6	5.0	0.2		
		MCWB	-2.9	-2.2	1.1	4.5	12.2	15.5	16.6	17.0	15.8	11.2	4.0	-0.3		
	5%	DB	-5.8	-3.6	1.8	7.1	18.2	23.3	24.2	22.3	18.1	11.2	2.9	-2.0		
		MCWB	-6.4	-4.7	0.0	3.1	10.5	14.5	16.0	16.0	14.3	9.1	1.9	-2.6		
	10%	DB	-8.7	-7.9	-0.4	4.9	15.0	21.0	22.1	20.3	15.8	8.9	1.3	-4.2		
		MCWB	-9.2	-8.7	-2.0	1.6	9.0	13.5	15.3	15.3	12.8	7.2	0.7	-4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	0.8	3.8	7.1	15.7	19.8	20.6	20.1	18.6	13.9	6.8	1.8		
		MCDB	0.9	2.1	6.1	11.9	21.7	26.8	25.1	23.8	22.5	16.6	7.3	2.3		
	2%	WB	-2.9	-2.1	1.6	4.8	13.4	17.8	18.5	18.3	16.8	11.6	4.3	-0.2		
		MCDB	-2.4	-1.1	3.2	8.6	19.8	22.8	22.7	22.0	19.8	13.2	4.9	0.2		
	5%	WB	-6.4	-4.8	-0.1	3.5	11.4	16.3	17.4	17.1	15.0	9.5	2.0	-2.6		
		MCDB	-5.9	-3.6	1.7	6.6	17.3	20.5	21.4	21.0	17.5	11.0	2.6	-2.1		
	10%	WB	-9.2	-8.8	-2.1	2.0	9.4	14.7	16.4	16.0	12.9	7.2	0.7	-4.7		
		MCDB	-8.7	-8.0	-0.3	4.5	14.5	19.1	20.2	19.3	15.3	8.6	1.3	-4.2		
(35) Mean Daily Temperature Range	MDBR		10.0	11.9	12.9	10.9	11.2	12.0	10.5	9.5	8.3	6.1	5.9	8.3		
	5% DB	MCDBR	10.7	10.9	11.7	12.1	17.6	16.0	14.9	13.3	12.1	10.1	6.7	7.1		
		MCWBR	10.5	9.9	9.6	8.0	9.5	7.8	7.0	6.8	7.9	7.8	6.3	6.9		
	5% WB	MCDBR	10.5	10.8	10.7	10.8	15.1	12.1	10.9	10.3	10.5	9.0	6.3	7.2		
		MCWBR	10.3	10.0	9.1	7.7	8.9	7.2	6.4	6.3	8.0	7.7	6.1	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.237	0.251	0.274	0.327	0.365	0.365	0.390	0.362	0.351	0.297	0.270	0.227		
	taud		2.205	2.227	2.201	2.145	2.307	2.386	2.358	2.441	2.437	2.534	2.247	2.186		
	Ebn at Noon		746	861	912	898	881	883	852	859	821	809	685	678		
	Edh at Noon		61	84	108	132	120	112	114	99	88	64	59	50		
(44) All-Sky Solar Radiation	RadAvg		1.01	2.19	3.81	5.28	5.42	5.49	4.76	3.85	2.68	1.47	0.64	0.58		
	RadStd		0.09	0.16	0.29	0.41	0.38	0.47	0.41	0.27	0.26	0.11	0.09	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	-0.76	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page